JENNIFER HSIA

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Machine Learning Ph.D. candidate working on LLM evaluation, model behavior, and reliability. Builds controlled evaluations that reveal when models reach conclusions their inputs don't justify, such as deferring to a user's unsupported claim or picking a plausible but wrong answer, and develops methods that reduce these failures.

EDUCATION

Carnegie Mellon University 2021–Present
Ph.D. in Machine Learning (expected May 2027) · M.S. in Computer Science (2023)
Advisors: Zachary C. Lipton and Aarti Singh

Princeton University 2017–2021
B.S.E. in Computer Science, magna cum laude · Minors in Statistics and Machine Learning

RESEARCH EXPERIENCE

Carnegie Mellon University | Graduate Researcher, Machine Learning Department 2021–Present
First or co-first author on all projects below.

Deference to user claims in LLM advice In submission
- Studying how users' claims about their own circumstances shape LLM advice.

Robust answer selection when retrieval favors a wrong answer EMNLP 2026
- Showed that LLMs can be pulled toward an answer that misses a key qualifier in the question (e.g., the film's Glinda when asked about Wicked's original Broadway cast), and, in controlled experiments, that adding passages supporting such answers makes these mistakes more likely.
- Developed and implemented Joint Constraint Aggregation (JCA), which extracts a question's requirements and compares candidate answers against them jointly, and built the evaluation pipeline over web-retrieved evidence and 3 open-weight models served with vLLM.
- JCA was the most accurate method in every setting tested, by 2–11 points over the best baseline, and lost less accuracy than any standard baseline when retrieval favored a wrong answer (3 evaluation settings × 3 instruction-tuned open-weight models).

Robustness of RAG systems to the number of retrieved passages ICML 2025
- Showed that evaluating RAG at a single retrieval depth hides brittleness: as more passages are added, some models peak early and then degrade, while others improve and then plateau.
- Introduced metrics for stability (how steady performance stays near the best depth) and scalability (how much a model gains from more passages), and found that which pattern a model follows tracked its sensitivity to distracting passages, holding across retrievers, rerankers, and prompts.
- Led the experimental design and analysis, and built the retrieval and reranking stack (BM25, dense retrievers with FAISS, cross-encoder rerankers).

Gaming explanation-faithfulness benchmarks Findings of EACL 2024
- Showed that ERASER, which scores explanations by re-running the model on partially masked inputs, can be gamed: a simple wrapper inflated scores sharply by detecting those inputs, without changing predictions or explanations on real inputs.
- For EVAL-X, encoding the predicted label in the explanation achieved the maximum score, though the benchmark was designed to block exactly this shortcut.

INDUSTRY EXPERIENCE

Meta | Research Scientist Intern, Ads Ranking May–August 2025
- Developed a compact representation of long user-behavior histories for ads ranking: older history is compressed while recent events are kept in full, to fit memory and serving-throughput limits.

- Trained ranking models with this representation; in offline evaluation on recent data, it improved normalized entropy (the standard offline ads-ranking metric) by 0.03% relative, above the internal threshold for a meaningful gain, at under 2% throughput cost.

PUBLICATIONS

** equal contribution*

When the Rule Is Clear but the Case Isn't: How User Claims Shape LLM Advice
Jennifer Hsia, Zachary C. Lipton, Aarti Singh. *In submission*

Joint Constraint Aggregation for Qualifier-Sensitive Retrieval-Augmented QA
Jennifer Hsia, Zachary C. Lipton, Aarti Singh. *EMNLP 2026* · Paper · Code

RAGGED: Towards Informed Design of Scalable and Stable RAG Systems
Jennifer Hsia*, Afreen Shaikh*, Zhiruo Wang, Graham Neubig. *ICML 2025* · *Earlier version: oral, NeurIPS 2024 Workshop on Adaptive Foundation Models* · Paper · Code

Goodhart's Law Applies to NLP's Explanation Benchmarks
Jennifer Hsia, Danish Pruthi, Aarti Singh, Zachary C. Lipton. *Findings of EACL 2024* · Paper · Code

An Experimental Study of Balance in Matrix Factorization
Jennifer Hsia, Peter Ramadge. *CISS 2021* · Paper

TECHNICAL SKILLS

Evaluation: LLM-as-judge design and validation against human labels; human annotation and inter-annotator agreement; statistical testing (bootstrap confidence intervals, paired tests)

Tools: Python, Hugging Face Transformers, vLLM, multi-GPU inference on Slurm, frontier-model APIs (OpenAI, Anthropic, Gemini), retrieval (BM25, dense retrievers, FAISS)

TEACHING & SERVICE

Teaching Assistant, CMU: Human-AI Complementarity for Decision Making (2025); Probabilistic Graphical Models (2023)

Machine Learning Department TA Award (one of four TAs recognized for 2021–2023) 2023

Workshop Organizer: Human Evaluation of Generative Models, NeurIPS 2022

Reviewer: ACL, EMNLP, ICML